

2024



Department of Premier and Cabinet Tasmania Achieves 100% Uptime Availability for Citizen-Facing Emergency Alerts Platform Using Multi-Region Deployments

Discover how the central agency of the Tasmanian State Government unified emergency communications, building on AWS to ensure safety and reliability during crises.

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100% availability and uptime since launch

using Active-Active multi-region



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532%

increased traffic managed during peak periods

Over 650,000

users engaged with the platform s



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Overview

The [Department of Premier and Cabinet \(DPAC\)](#), the central agency of the Tasmanian State Government, delivers public policy, government information, and services to enhance the wellbeing of over 500,000 residents. This includes delivering real-time, accurate emergency updates and warnings to its citizens during events, such as during bushfires and floods, through the TasALERT service – a centralized, online platform built on [Amazon Web Services \(AWS\)](#).

TasALERT has achieved 100% availability and uptime since launch using AWS’s robust multi-region infrastructure available in Australia ensuring emergency alerts can reach citizens when it matters most. DPAC was the first Australian public agency to deploy an Active-Active multi-region design with complete workload infrastructure and data across both the Sydney and Melbourne Regions, setting a new standard for public safety communication in Australia.

DPAC worked closely with AWS partner Ionata Digital to design the TasALERT platform to be highly scalable and reliable. For instance, during the February 2024 bushfires, the platform managed a 532 percent increase in traffic, maintaining continuous availability and quick response times. With automated scaling, DPAC can now focus on managing emergencies rather than infrastructure.



Opportunity | Centralizing Emergency Alerts to Enhance Public Safety

Tasmania frequently experiences emergencies such as bushfires and floods, requiring a reliable system for disseminating emergency alerts.

However, the delivery of emergency information in Tasmania was previously fragmented across multiple agencies, including the Tasmanian Fire Service, Tasmania Police, and the State Emergency Service. This made it difficult for residents to find reliable information promptly during emergencies.

In 2013, DPAC launched the TasALERT platform to consolidate information from different sources, aiming to provide credible and timely notifications to its residents. However, the platform mainly republished alerts from other sources, causing potential delays in crucial information dissemination. Additionally, the legacy platform lacked the robust infrastructure required to ensure continuous operation during regional outages, creating information gaps if one system failed.

DPAC identified the need for a centralized platform that could deliver near real-time alerts. The new platform required high availability, reliability, and scalability to ensure the highest possible uptime and handle traffic surges during emergencies. In 2021, the agency decided to redesign the TasALERT platform to improve its emergency response capabilities and enhance public safety for all residents.

Solution | Building a Responsive and Resilient Emergency Communications Platform

Recognizing the importance of a reliable and scalable emergency alert system, DPAC engaged AWS Partner Ionata Digital to overhaul the TasALERT platform. The new system now delivers near real-time emergency updates, providing Tasmanian residents with timely emergency information.

For high availability and resiliency, TasALERT leverages AWS's multi-region infrastructure in Australia, including the newly launched Melbourne Region in January 2023, alongside the Asia Pacific (Sydney) Region. Each region comprises three Availability Zones, each with independent power, cooling, and physical security, connected by redundant, ultra-low latency networks. To ensure maximum fault tolerance active/active hosting was implemented for TasALERT across both regions. This approach, a first for an Australian public agency, helps enable uninterrupted operations and resilience, even during rare regional service impairments, while ensuring all data remains securely stored within Australia.

TasALERT leverages several AWS services to ensure high performance, scalability, and reliability. [AWS Elastic Beanstalk](#) automatically scales resources to handle traffic surges, while [AWS Lambda](#) processes incoming alerts in real-time. [Amazon CloudFront](#) accelerates content delivery, and [Amazon Simple Storage Service \(S3\)](#) efficiently manages static content, improving load times and user experience. [Amazon API Gateway](#) scales to handle high volumes of concurrent API calls from various emergency services, while [Amazon DynamoDB](#) provides reliable data storage for real-time alerts. Through this combination of services, TasALERT can



availability, even during high-traffic emergency events, ensuring Tasmanian residents receive critical information quickly and reliably.

Outcome | Transforming Emergency Communications for Tasmania's Residents

The successful transformation of the TasALERT platform on AWS has revolutionized DPAC's emergency management capabilities. This has set a new standard for public safety communication in Australia. Over 650,000 users visited TasALERT in the year leading up to June 2024, with an average engagement time of 2.5 minutes per visit. The platform has become the central hub for emergency communications in Tasmania, providing real-time, life-saving information to the residents. The accessibility to critical updates not only helps in immediate preparations but also provides a sense of security and reassurance to the community, knowing they are informed and able to respond effectively to any crisis.

“It was essential that the hosting solution for TasALERT can withstand huge spikes in usage by members of the public in the event of a bushfire or flood” said Martin Anderson, CEO and Founder, Ionata Digital. “The fact that the TasALERT platform has achieved 100% availability and uptime since launch, including during a number of emergency events, is a testament to the strength of the multi-region cloud architecture that we can deliver with AWS.”

The redesigned TasALERT platform now consolidates alerts from various services and delivers real-time updates during crises such as floods, bushfires, extreme weather and even cyber security events, ensuring Tasmanians are informed and well-prepared during emergencies. The platform's robust performance during recent natural disasters has demonstrated its effectiveness in keeping the community safe.

Building on this success, DPAC has recently launched a companion mobile application for TasALERT. This app introduced an innovative feature called WatchZones, further extending the reach of emergency alerts and making critical information even more accessible to Tasmanian residents. This initiative enhances the state's emergency response capabilities, ensuring that vital information is delivered to residents wherever they are in real time.

“Using AWS has increased our confidence in providing timely emergency information and warnings to the community. The multi-jurisdictional infrastructure provides us with the comfort that our platform will reliably deliver important information the community needs, particularly during emergencies” – Todd Crawford, Executive Director, Resilience and Recovery Tasmania.

Learn More

To learn more, visit aws.amazon.com/government-education/government.

About the Department of Premier and Cabinet (DPAC) Tasmania

The Department of Premier and Cabinet (DPAC) is the central agency of the Tasmanian State Government. The Department provides a broad range of services to Cabinet, other members of Parliament, Government Ministers and the community. The Department works closely with the public sector, the community, local government, the Australian Government and other state and territory governments.

AWS Services Used

AWS Elastic Beanstalk

Elastic Beanstalk is a service for deploying and scaling web applications and services. Upload your code and Elastic Beanstalk automatically handles the deployment—from capacity provisioning, load balancing, and auto scaling to application health monitoring.

[Learn more »](#)

AWS Lambda

AWS Lambda is a compute service that runs your code in response to events and automatically manages the compute resources, making it the fastest way to turn an idea into a modern, production, serverless applications.

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Amazon API Gateway

Amazon API Gateway is a fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure APIs at any scale. APIs act as the "front door" for applications to access data, business logic, or functionality from your backend services.

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Amazon Simple Storage Service (Amazon S3)

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

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